Derivations: PGI-RC Correction

Notation

Symbol	Definition
σ^2_{tot}	Total variance in the outcome: $\sigma^2(\tau_j) + \sigma^2(\delta_{ij})$ (from M0)
$\sigma^2(\tau_j)$	Between-family variance from M0 (intercept only)
$\sigma^2(\delta_{ij})$	Within-family (residual) variance from M0
$\sigma^2(\tau_j^*)$	Between-family variance from M1 (conditioning on PGI)
$\sigma^2(\delta_{ij}^*)$	Within-family variance from M1 (conditioning on PGI)
$\sigma^2(\tau_j')$	Between-family variance from M2 (conditioning on PGI + ascribed)
$\sigma^2(\delta_{ij}')$	Within-family variance from M2 (conditioning on PGI + ascribed)
w	$[\sigma^2(\delta_{ij}^*) - \sigma^2(\delta_{ij}')] \,/\, \sigma^2_{\text{tot}}$
Sibcorr	$\sigma^2(\tau_j) \,/\, \sigma^2_{\text{tot}}$ — sibling correlation (from M0, unaffected by PGI noise)
condcorr	$\sigma^2(\tau_j^*) \,/\, \sigma^2_{\text{tot}}$
Δ_{between}	$[\sigma^2(\tau_j) - \sigma^2(\tau_j^*)] \,/\, \sigma^2_{\text{tot}}$ — observed between-family genetic share
Δ_{within}	$[\sigma^2(\delta_{ij}) - \sigma^2(\delta_{ij}^*)] \,/\, \sigma^2_{\text{tot}}$ — observed within-family genetic share
ρ_{between}	$\sqrt{h^2_{\text{between}} \,/\, (\text{ICC} \times R^2_{\text{between}})} \geq 1$
ρ_{within}	$\sqrt{h^2_{\text{within}} \,/\, ((1 - \text{ICC}) \times R^2_{\text{within}})} \geq 1$
g^*	True additive SNP factor; PGI $= (g^* + e)/\rho$ where e is uncorrelated noise

Setup

The PGI is a noisy proxy for the true genetic factor:

$$\text{PGI} = \frac{g^* + e}{\rho}$$

where e is estimation error uncorrelated with g^* and with all other variables. M1 therefore absorbs less variance than it would with the true g^*. The observed genetic shares are:

$$\Delta_{\text{between}} = \frac{\sigma^2(\tau_j) - \sigma^2(\tau_j^*)}{\sigma^2_{\text{tot}}}, \qquad \Delta_{\text{within}} = \frac{\sigma^2(\delta_{ij}) - \sigma^2(\delta_{ij}^*)}{\sigma^2_{\text{tot}}}$$

Conditioning on g^* instead of PGI absorbs ρ^2 times more variance at each level:

$$\sigma^2(\tau_j^*)_{\text{true}} = \sigma^2(\tau_j) - \rho^2_{\text{between}}\,\Delta_{\text{between}}\,\sigma^2_{\text{tot}}$$

$$\sigma^2(\delta_{ij}^*)_{\text{true}} = \sigma^2(\delta_{ij}) - \rho^2_{\text{within}}\,\Delta_{\text{within}}\,\sigma^2_{\text{tot}}$$

For M2, we approximate that the ascribed-characteristics contribution is unaffected by PGI noise (reasonable when PGI and ascribed variables are weakly correlated):

$$\sigma^2(\delta_{ij}^*)_{\text{true}} - \sigma^2(\delta'_{ij})_{\text{true}} \approx \sigma^2(\delta_{ij}^*) - \sigma^2(\delta'_{ij}) = w \cdot \sigma^2_{\text{tot}}$$

Corrected IOLib

Start from the definition (Eq. 4 in the analytical strategy):

$$\text{IOLib} = \frac{\sigma^2(\tau_j^*)}{\sigma^2_{\text{tot}}} + \frac{\sigma^2(\delta_{ij}^*) - \sigma^2(\delta'_{ij})}{\sigma^2_{\text{tot}}} = \text{condcorr} + w$$

Replace $\sigma^2(\tau_j^*)$ with the corrected value. From the definition of Δ_{between}:

$$\sigma^2(\tau_j^*) = \sigma^2(\tau_j) - \Delta_{\text{between}}\,\sigma^2_{\text{tot}}$$

so the true g^* would absorb ρ^2_{between} times as much between-family variance:

$$\sigma^2(\tau_j^*)_{\text{true}} = \sigma^2(\tau_j) - \rho^2_{\text{between}}\,\Delta_{\text{between}}\,\sigma^2_{\text{tot}}$$

Dividing by σ^2_{tot}:

$$\frac{\sigma^2(\tau_j^*)_{\text{true}}}{\sigma^2_{\text{tot}}} = \text{Sibcorr} - \rho^2_{\text{between}}\,\Delta_{\text{between}}$$

Using the M2 approximation (w is unchanged) and noting that condcorr $=$ Sibcorr $- \Delta_{\text{between}}$:

$$\text{IOLib}_{\text{RC}} = \text{Sibcorr} - \rho^2_{\text{between}}\,\Delta_{\text{between}} + w = \text{condcorr} + \Delta_{\text{between}} - \rho^2_{\text{between}}\,\Delta_{\text{between}} + w$$

$$\boxed{\text{IOLib}_{\text{RC}} = \text{IOLib} - (\rho^2_{\text{between}} - 1)\,\Delta_{\text{between}}}$$

Since $\rho_{\text{between}} > 1$, IOLib decreases: the noisy PGI left too much variance in $\sigma^2(\tau_j^*)$, over-attributing shared-family-environment variance to the liberal measure.

Corrected IORad

Sibcorr $= \sigma^2(\tau_j)/\sigma^2_{\text{tot}}$ comes from M0 (no predictors) and is unaffected by PGI noise. The bias enters through $\sigma^2(\delta'_{ij})_{\text{true}}$ in M2. Using the M2 approximation:

$$\sigma^2(\delta'_{ij})_{\text{true}} = \sigma^2(\delta^*_{ij})_{\text{true}} - [\sigma^2(\delta^*_{ij}) - \sigma^2(\delta'_{ij})]$$

The corrected within-family variance explained by natural talents and ascribed characteristics is:

$$\sigma^2(\delta_{ij}) - \sigma^2(\delta'_{ij})_{\text{true}} = \rho^2_{\text{within}}\,\Delta_{\text{within}}\,\sigma^2_{\text{tot}} + [\sigma^2(\delta^*_{ij}) - \sigma^2(\delta'_{ij})]$$

Dividing by σ^2_{tot} and adding Sibcorr:

$$\boxed{\text{IORad}_{\text{RC}} = \text{Sibcorr} + \rho^2_{\text{within}}\,\Delta_{\text{within}} + w}$$

Since $\rho_{\text{within}} > 1$, IORad increases: the within-family genetic contribution was underestimated because the noisy PGI absorbed less within-family variance than the true g^* would.

Verification

The corrected gap is ρ^2 times each genetic share:

$$\text{IORad}_{\text{RC}} - \text{IOLib}_{\text{RC}} = \rho^2_{\text{between}}\,\Delta_{\text{between}} + \rho^2_{\text{within}}\,\Delta_{\text{within}}$$

Scale correction for ρ

h^2_{between} and h^2_{within} from the literature are fractions of **total** phenotypic variance. R^2_{between} (from the family-means regression) and R^2_{within} (from the within-family demeaned regression) are fractions of **between-** and **within-family** variance respectively. Dividing directly mixes scales. The ICC converts each R^2 to the total-variance scale:

$$\rho_{\text{between}} = \sqrt{\frac{h^2_{\text{between}}}{\text{ICC} \times R^2_{\text{between}}}}, \qquad \rho_{\text{within}} = \sqrt{\frac{h^2_{\text{within}}}{(1-\text{ICC}) \times R^2_{\text{within}}}}$$

This is consistent with the additive decomposition of R^2_{pop}:

$$R^2_{\text{pop}} = \text{ICC} \times R^2_{\text{between}} + (1-\text{ICC}) \times R^2_{\text{within}}$$

which differs from the additive h^2 decomposition ($h^2_{\text{pop}} = h^2_{\text{between}} + h^2_{\text{within}}$) precisely because R^2 components use level-specific denominators while h^2 components all use total variance as the denominator.